ELVIRA NURGALIEVA

Colorado Engineer Intern (EI.0081521) | Industrial & Manufacturing Engineering

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PROFESSIONAL SUMMARY

Colorado Engineer Intern who passed the FE Industrial & Systems exam, with 3.5 years of experience in 3D scanning, reverse engineering, and technical workflow analysis, plus operations coordination and high-volume retail operations. Focused on process improvement, quality, root-cause analysis, and technical documentation. U.S. permanent resident; no sponsorship required.

CORE QUALIFICATIONS

Process improvement and workflow analysis | Requirements analysis | Root-cause analysis, 5 Whys, fishbone analysis | Lean and DMAIC concepts | Process mapping and standard work | Reverse engineering and scan-to-CAD | Technical documentation | Trade-off analysis | Stakeholder communication and technical training | Excel-based analysis

PROFESSIONAL EXPERIENCE

Sales Associate, Paint Department

Apr 2024 - Present

The Home Depot | Castle Rock, Colorado

- Process customer and contractor paint orders through tinting and mixing equipment to formula and quality standards in a high-volume store, following established safety and operating procedures.
- Translate project requirements (surface, environment, coverage, budget) into product and process recommendations; coordinate practical solutions when standard approaches do not fit the need.
- Apply Gemba observation, process mapping, 5 Whys, fishbone analysis, and future-state thinking in an independent Kaizen study of shelving and replenishment flow, with attention to bottlenecks, rework, and standard work.

Relocation to the U.S. and Engineering Upskilling

Mar 2022 - Mar 2024

Career break

- Completed Georgia Institute of Technology engineering mechanics coursework; continued CAD and reverse-engineering practice and studied industrial and systems engineering topics toward the FE exam.

Sales Engineer - 3D Scanning and Reverse Engineering

Aug 2018 - Feb 2022

Thor3D | Moscow, Russia

- Led structured technical discovery with engineering and manufacturing customers to clarify objectives, constraints, workflow pain points, software compatibility, accuracy needs, and implementation requirements.
- Mapped and evaluated workflows across equipment, software, data processing, reverse engineering, CAD, inspection, and user methods; identified bottlenecks, failure points, and opportunities to improve repeatability and data quality.
- Diagnosed technical and workflow issues, documented findings, coordinated corrective guidance, and performed trade-off and feasibility analyses balancing functionality, accuracy, cost, implementation effort, operational risk, and user adoption.
- Created CAD and scan-derived models using SolidWorks, Fusion 360, Geomagic Design X, Rhinoceros, and Blender; supported adoption of digital engineering workflows.

Dispatch Coordinator

Dec 2017 - Jul 2018

MultiMedia Communications Ltd | Christchurch, New Zealand

- Coordinated technicians, subcontractors, suppliers, schedules, materials, and project documentation for telecommunications infrastructure installations.
- Monitored execution, resolved scheduling and resource conflicts, maintained operational records, and escalated issues affecting service commitments and project delivery.

SELECTED ENGINEERING PROJECTS

Mesh-to-CAD Reverse Engineering Workflow (Thor3D) - Converted 3D scan data into engineering-ready CAD models and drawings for legacy parts, wear analysis, and quality control (Geomagic Design X, SolidWorks, Fusion 360).

Aircraft Fuselage Frame Process Redesign (Bauman final project) - Redesigned the process for Tupolev fuselage frames, replacing pressure die casting with CNC milling for tighter dimensional control, and evaluated the economics at production volume.

Retail Material Flow and Kaizen Study - Current-state mapping, waste identification, 5 Whys, fishbone analysis, and future-state design for a recurring replenishment and material-flow problem.

CREDENTIALS AND EDUCATION

Colorado Engineer Intern, active - EI.0081521 (July 2026) | **Fundamentals of Engineering**, Industrial & Systems - Passed (June 2026)

Bauman Moscow State Technical University - Engineer's Degree (Specialist), Systems Engineering; NCEES evaluated as equivalent to a U.S. engineering bachelor's degree.

Georgia Institute of Technology - Engineering Mechanics coursework and certificates | **Southern Institute of Technology** - Postgraduate Diploma, Business Enterprise / Management

TECHNICAL TOOLS AND PROFESSIONAL DEVELOPMENT

Tools: SolidWorks, Fusion 360, Geomagic Design X, Rhinoceros, Blender, Excel, Word, PowerPoint, Outlook, Google Workspace, Git, GitHub.

In progress: Lean Six Sigma Green Belt study, Python for engineering and data analysis, and Power BI. Uses generative-AI tools to structure analysis and improve technical documentation.